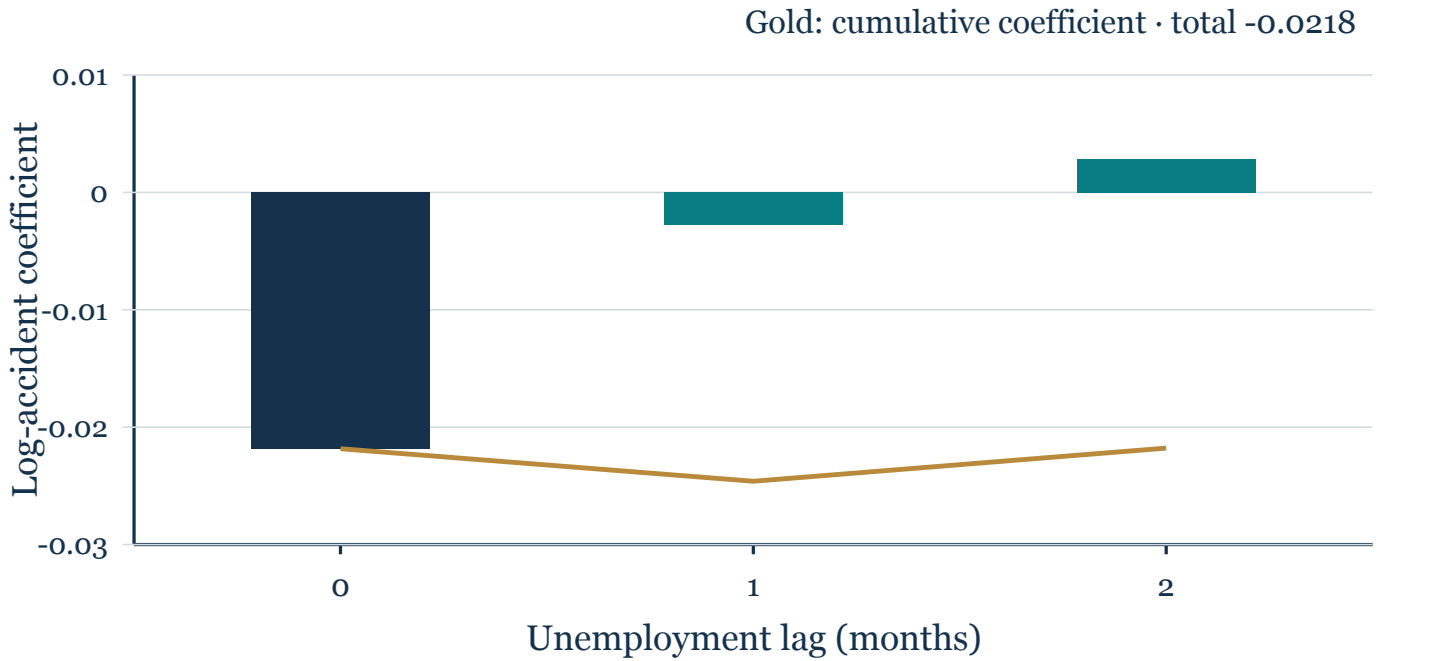
Does unemployment affect accidents only this month?

Allow a relationship to unfold over several periods.



Actual monthly unemployment coefficients from the finite distributed-lag regression. Cumulative means the sum through the selected lag.

Define lags, differences and horizons

They are different operations on ordered observations.

> $LX_t = X_{t-1}$

> $\Delta X_t = X_t - X_{t-1}$

Write a finite distributed-lag model

Every coefficient has an outcome date and a regressor date.

$$Y_t = a + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \text{controls} + u_t$$

Trace a one-period regressor change

Its outcome responses occur at several dates.

> Horizon h: $0, 1, 2, 3$

> Temporary-input responses: $0.20, 0.10, -0.05, 0$

> Cumulative response: $0.20 + 0.10 - 0.05 = 0.25$

A permanent change has another time path

The long-run level effect reaches the lag sum in this model.

> Permanent-input responses: $0.20, 0.30, 0.25, 0.25, \ldots$

> Temporary cumulative response and permanent long-run level contrast are different quantities.

Lag sum

Try the comparison before opening the explanation.

Practice

A finite distributed-lag model has coefficients 0.2, 0.1 and −0.05 on current x, lag 1 and lag 2. Compute the impact and the cumulative response to a one-period unit x shock.

Lag sum: explanation

Follow the calculation and interpret its units.

> Impact is 0.2. The three-period cumulative response is $0.2 + 0.1 - 0.05 = 0.25$ outcome units. In this finite-lag model, a permanent unit x increase has a long-run level effect of 0.25 as well. An autoregressive dependent-variable model has a different multiplier formula.

Introduce the uncertainty option before reading the lag table

Newey–West changes the covariance estimate around an unchanged OLS fit.

> Newey–West/HAC: allow suitable unequal spread and time dependence in uncertainty estimation.

> Four covariance lags here mean four months; the mean equation is unchanged.

Read the observed lag estimates

Let the application determine the pattern.

```
. newey logacc unem L(1/2).unem trend i.month, lag(4)

------------------------------------------------------------------------------
             |             Newey–West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        unem |
         --. |  -.0218345   .0073752    -2.96   0.004    -.0364866   -.0071823
         L1. |  -.0027711   .0075824    -0.37   0.716    -.0178349    .0122927
         L2. |   .0028242   .0079337     0.36   0.723    -.0129375    .0185858
             |
       trend |   .0018359   .0002514     7.30   0.000     .0013365    .0023352
             |
       _cons |   10.67207   .0367848   290.12   0.000     10.59899    10.74515
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Estimate uncertainty for the lag sum

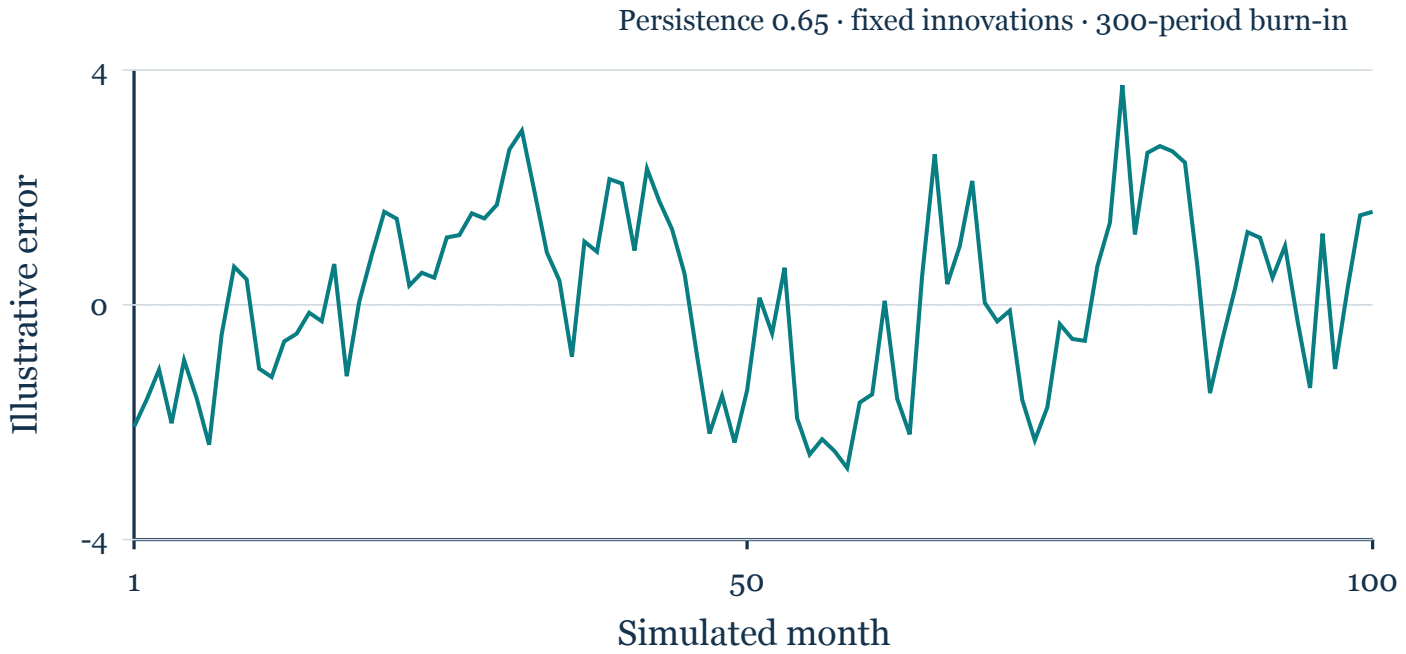
The coefficient covariances matter.

```
. lincom unem+L.unem+L2.unem

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |  -.0217814    .004154    -5.24   0.000    -.0300341   -.0135287
------------------------------------------------------------------------------
```

An AR(1) model carries the outcome forward

The persistence parameter determines whether shocks fade.



Illustrative simulation · fixed innovations · vertical range adapts to the selected persistence

$Y_t = a + \rho Y_{t-1} + \varepsilon_t$

One innovation with $\rho = 0.6$: $1, 0.6, 0.36, 0.216, \ldots$

A unit root does not let shocks fade in the usual way

A random walk accumulates innovations.

> Random walk: $Y_t = Y_{t-1} + \varepsilon_t$

> If $Y_0 = 0$, $\mathrm{Var}(Y_t) = t\sigma^2$

> $\Delta Y_t = \varepsilon_t$

Stationarity describes stable statistical features

Specify which definition is being used.

> Covariance stationary: stable finite mean/variance and lag-dependent covariance.

> AR(1) stationary regime requires $|\rho| < 1$ and suitable innovations/initialization.

> Strict stationarity concerns full joint distributions.

A permanent input change can be amplified by outcome persistence

Solve for the steady difference instead of adding coefficients blindly.

> $\Delta Y_h = 0.6\Delta Y_{h-1} + 0.2: \quad 0.20, 0.32, 0.392, \ldots$

> $\Delta Y = 0.6\Delta Y + 0.2 \Rightarrow \Delta Y = 0.5$

> Long-run multiplier: $\beta/(1-\rho)$ for a stable recursion.

AR long run

Try the comparison before opening the explanation.

Practice

For $Y_t = a + 0.6Y_{t-1} + 0.2X_t + e_t$, what is the long-run response to a permanent unit increase in x under a stable model?

AR long run: explanation

Follow the calculation and interpret its units.

> The steady-state change solves $\Delta Y = 0.6\Delta Y + 0.2$, so $\Delta Y = 0.2/(1 - 0.6) = 0.5$. This calculation assumes a stable dynamic relation and an exogenous permanent x change with other determinants fixed.

Inspect persistence in the accident model

The fitted lag is a conditional association.

```
. newey logacc L.logacc trend i.month, lag(4)

------------------------------------------------------------------------------
             |             Newey–West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      logacc |
         L1. |   .6783359   .0825587     8.22   0.000     .5143906    .8422811
             |
       trend |   .0009218   .0002548     3.62   0.000     .0004158    .0014278
             |
       _cons |   3.276324   .8753799     3.74   0.000     1.537993    5.014655
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

Differencing changes the target relationship

It can remove an accumulating level component in a specific model.

> For the simple random walk, first differencing leaves the innovation.

> A level relationship and a relationship between changes answer different empirical questions.

Unrelated trending series can appear related

A high R-squared can be misleading.

> Two independent series with accumulating random shocks can produce a striking regression fit in levels even without an economic relationship.

> Inspect time plots and model the source of persistence before interpreting a slope.

Trending regression

Try the comparison before opening the explanation.

Practice

Two unrelated series both rise over time and produce a high R^2 when regressed on each other. What should you do before interpreting the slope?

Trending regression: explanation

Follow the calculation and interpret its units.

> Inspect their time plots and consider deterministic trends, stochastic trends and stationarity. A high R^2 does not establish a meaningful relationship. Choose a specification and inference procedure justified by the series' properties; differencing every series automatically is also not a substitute for reasoning.

Weak dependence concerns information far apart in time

Persistence and large-sample reasoning must be compatible.

> Weak dependence: sufficiently distant information becomes weak enough for the stated limit result.

> Stationarity alone does not guarantee that time averages behave like independent draws.

The sample changes when lags are added

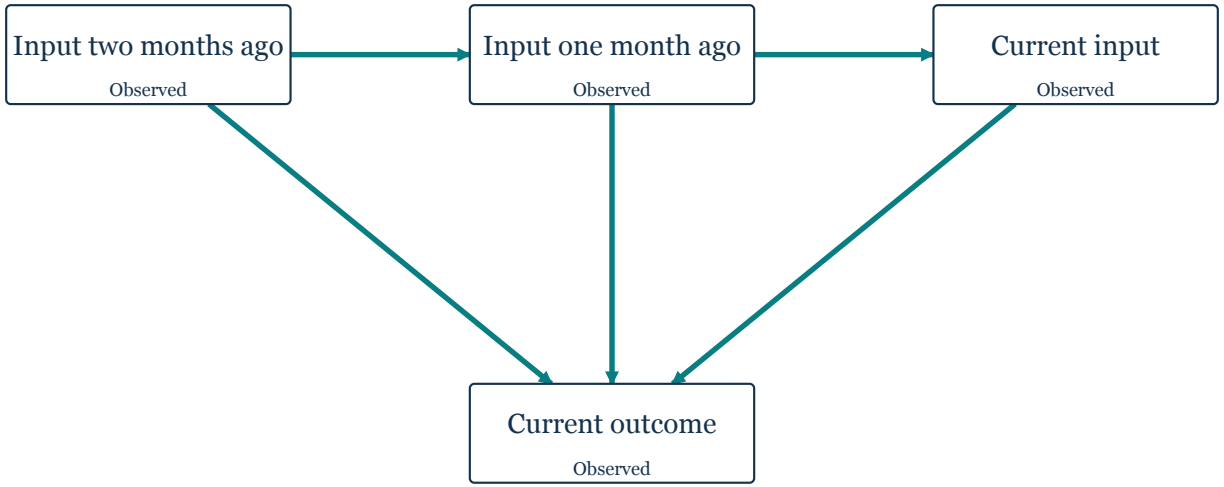
Compare fit and tests on compatible observations.

> A two-lag model loses the first two observations if no earlier data are available.

> Define the common estimation window before comparing models or applying a fit-loss test.

Explain a dynamic result with a time path

One label such as effect is not enough.

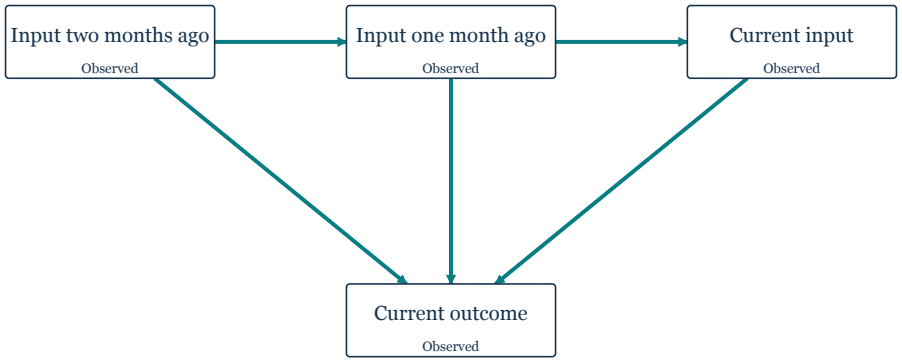


3 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

DAG practice: partial lag effect or propagated input change?

Trace the routes before choosing a control.



3 open directed routes · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

An earlier input influences the outcome directly and by changing later input values. Which routes remain if the later inputs are held fixed? Why can the response to a naturally propagated input innovation differ from the coefficient on that earlier input?

DAG practice: partial lag effect or propagated input change?: explanation

Explain what conditioning changes in this stated graph.

> Holding both later inputs fixed blocks the routes through those input nodes. The direct earlier-input-to-current-outcome route remains. Its partial contrast differs from an intervention that also changes later values of the input process. For a propagated innovation, trace the resulting entire input path and combine it with the distributed-lag coefficients. For a one-period imposed regressor increase, specify that all other dates remain fixed. A sum of lag coefficients is meaningful for the stated path and model; the word shock alone does not define which path was changed.

Trace the actual monthly lag pattern

Reproduce the evidence, then explain its economic meaning.

Practice

Extend unemployment lags from two to three months. First restrict both fits to the same 105 available months. Compare the cumulative coefficients and their lincom SEs. Keep your commands, log, output and interpretation.

Trace the actual monthly lag pattern: explanation

A reproducible result needs both a calculation and an interpretation.

> Common 105 months: two-lag sum −0.018658; three-lag sum −0.018684.

> Corresponding robust SEs: 0.003752 and 0.003761.

> Fixing the sample isolates the change in lag specification.

Next, examine what the fitted errors still remember

Dynamics and uncertainty are related but distinct.

> We have defined stationary behaviour, distributed lags and autoregressive persistence, and distinguished several dynamic contrasts.

> The next lesson studies serial correlation in what the model leaves unexplained.